

Arduino-Based IoT Project Proposal Template

1. Project Title

[Enter a clear and specific project title]

2. Proponents

- Names: [Group members]
- Course/Section: [Course and section]
- Instructor: [Instructor's name]
- Date: [Submission date]

3. Background of the Project

Briefly describe the problem or need that the project will address. Explain why an Arduino-based IoT solution is appropriate.

4. Project Objectives

The project aims to:

1. Design and develop [main purpose of the prototype].
2. Use sensors to collect [type of data].
3. Transmit or display the collected data through [IoT platform/application].
4. Evaluate the prototype's functionality, accuracy, and reliability.

5. Proposed Solution

Describe how the proposed project will work using the following flow:

Input/Sensor → Arduino Processing → Internet Connection → IoT Platform → User Notification or Action

6. Target Users

The intended users are **[identify users]**. They will use the project to **[state its main benefit]**.

7. Hardware and Software Requirements

Category	Proposed Components
Microcontroller	Arduino Uno, Nano, ESP32, or other board
Sensors	[List required sensors]
Outputs	LED, buzzer, relay, motor, display, etc.
Connectivity	Wi-Fi, Bluetooth, GSM, or Ethernet
IoT Platform	Blynk, ThingSpeak, Firebase, Arduino IoT Cloud, etc.
Development Software	Arduino IDE and related applications

8. Expected Output

A functional Arduino-based IoT project capable of:

- Monitoring [**specific condition**];
- Sending data through the internet;
- Displaying real-time or recorded information; and
- Providing alerts or automated actions.

9. Testing and Evaluation

The project will be evaluated according to:

- Functionality
- Sensor accuracy
- Data-transmission reliability
- Response time
- Ease of use
- Electrical and operational safety

10. Work Plan

Activity	Target Schedule
Planning and component selection	Week 1
Circuit and system design	Week 2
Programming and assembly	Weeks 3–4
IoT platform integration	Week 5
Testing and improvement	Week 6
Documentation and presentation	Week 7

11. Estimated Budget

Component	Quantity	Estimated Cost
[Component]	[Qty.]	₱[Amount]
[Component]	[Qty.]	₱[Amount]
Total		₱[Total]

12. Limitations

The project is limited to [location, users, sensor range, connectivity, power source, or testing conditions].

13. References

List the books, journal articles, technical documentation, and credible websites used in developing the proposal.

Rubric for Arduino-Based IoT Project Proposal Presentation

Criteria	Weight	Excellent	Proficient	Developing	Beginning
Problem and Project Relevance	15%	Clearly explains a significant problem and presents a highly relevant IoT solution.	Explains the problem and provides a relevant solution.	Problem or relevance is partially explained.	Problem is unclear or unrelated to the proposed solution.
Objectives and Expected Output	15%	Objectives are specific, measurable, achievable, and fully aligned with the expected output.	Objectives are clear and generally aligned with the output.	Some objectives are vague or difficult to measure.	Objectives are unclear, unrealistic, or unrelated to the output.
Arduino and IoT System Design	20%	Clearly presents the complete input–process–output and IoT data flow, including communication and user access.	Presents an understandable system design with minor missing details.	System design is incomplete or contains unclear connections.	System design is missing or technically incorrect.
Components and Technology Selection	15%	Hardware, sensors, connectivity, software, and IoT platform are appropriate and properly justified.	Most technologies are suitable, with adequate explanation.	Some components are unsuitable or insufficiently explained.	Components are inappropriate, incomplete, or unidentified.
Feasibility, Budget, and Work Plan	15%	Project is realistic, affordable, safe, and achievable within the proposed schedule.	Project is generally feasible, with minor concerns.	Project has notable concerns regarding cost, time, resources, or safety.	Project is impractical or lacks a workable plan.

Testing and Evaluation Plan	10%	Provides clear and appropriate methods for testing functionality, accuracy, reliability, and usability.	Provides a suitable testing plan with minor omissions.	Testing methods are general or incomplete.	Testing and evaluation procedures are missing or inappropriate.
Presentation and Visual Organization	5%	Presentation is clear, concise, well-organized, and supported by readable visuals or diagrams.	Presentation is organized and understandable.	Presentation contains unclear, crowded, or poorly organized information.	Presentation is difficult to understand or lacks organization.
Team Participation and Question Handling	5%	Members participate meaningfully and answer questions accurately and confidently.	Most members participate and answer questions adequately.	Participation is uneven or responses show limited understanding.	Minimal participation or inability to answer basic questions.
Total	100%				

Performance Scale

Level	Score
Excellent	4
Proficient	3
Developing	2
Beginning	1

Computing the Score

For each criterion:

$$\text{Criterion Score} = \frac{\text{Rating}}{4} \times \text{Weight}$$

Example: If the group receives a rating of **3** for *Arduino and IoT System Design*:

$$\frac{3}{4} \times 20 = 15 \text{ points}$$